

Piecewise Rules for Rates and Fees

Answer Key

Algebra 1

Quick Answers

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|--------|--------|--------------------|
| 1. 13 | 5. 3.5 | 9. $t = 4, y = 10$ |
| 2. 11 | 6. 78 | 10. 0.16 |
| 3. 103 | 7. 10 | |
| 4. 7 | 8. — | |
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Curriculum

Level: Algebra 1

HSA-CED.A.1–A.4 – *problems 2, 3, 4, 5, 9, 10*

HSF-IF.A–HSF-IF.C – *problems 1, 6, 7, 8*

Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

3. *If they got 128.0: billed every kilowatt-hour at the second-tier rate instead of only the ones above the 500 kWh allowance*

1. City garage: \$4.00 up to 2 hours, \$3.00 per additional hour, up to 8 hours. Find $C(5)$.

The flat fee covers everything up to the breakpoint at $t = 2$; past it, only the hours *beyond* the first 2 are charged, which is $t - 2$ of them.

$$C(t) = \begin{cases} 4.00, & 0 < t \leq 2 \\ 4.00 + 3.00(t - 2), & 2 < t \leq 8 \end{cases}$$

A 5-hour stay satisfies $2 < 5 \leq 8$, so use the second branch: $C(5) = 4.00 + 3.00(5 - 2) = 4.00 + 3.00 \cdot 3 = 4.00 + 9.00$ dollars.

$C(5) = 13.00$

2. Shipper: \$6.00 up to 3 pounds, then \$1.25 per pound over 3. Find $C(7)$.

Same shape, different units. The charged quantity past the breakpoint is $w - 3$ pounds.

$$C(w) = \begin{cases} 6.00, & 0 < w \leq 3 \\ 6.00 + 1.25(w - 3), & w > 3 \end{cases}$$

A 7-pound package uses the second branch: $7 - 3 = 4$ pounds are charged at \$1.25 each, so $C(7) = 6.00 + 1.25 \cdot 4 = 6.00 + 5.00$ dollars.

$C(7) = 11.00$

3. Utility: 500 kilowatt-hours at \$0.11, everything above 500 at \$0.16. Priya used 800 and claims \$128.00.

$$C(n) = \begin{cases} 0.11n, & 0 \leq n \leq 500 \\ 0.11(500) + 0.16(n - 500), & n > 500 \end{cases}$$

Priya's mistake: she applied the second-tier rate to the whole month, computing $0.16 \times 800 = 128.00$. The higher rate applies only to the kilowatt-hours *above* the 500 allowance; the first 500 always bill at \$0.11.

Correct bill: the first 500 kilowatt-hours cost $0.11 \times 500 = 55.00$ dollars. The remaining $800 - 500 = 300$ kilowatt-hours cost $0.16 \times 300 = 48.00$ dollars. Total: $55.00 + 48.00$ dollars.

$$C(800) = 103.00$$

4. Taxi: \$3.50 for the first mile, \$2.25 per additional mile. Find the trip length whose fare is \$17.00.

$$C(m) = \begin{cases} 3.50, & 0 < m \leq 1 \\ 3.50 + 2.25(m - 1), & m > 1 \end{cases}$$

A \$17.00 fare is well past the \$3.50 minimum, so it lives on the second branch. Set that branch equal to the fare and solve:

$$3.50 + 2.25(m - 1) = 17.00$$

$$2.25(m - 1) = 13.50$$

$$m - 1 = 6$$

$$m = 7$$

Check: $3.50 + 2.25 \cdot 6 = 3.50 + 13.50 = 17.00$ dollars. The trip was 7 miles.

$$m = 7 \text{ mi}$$

5. Pool: \$8.00 for the first 3 hours, k dollars per additional hour; a 7-hour visit costs \$22.00. Find k .

$$C(t) = \begin{cases} 8.00, & 0 < t \leq 3 \\ 8.00 + k(t - 3), & t > 3 \end{cases}$$

A 7-hour visit is charged for $7 - 3 = 4$ extra hours:

$$8.00 + k(7 - 3) = 22.00$$

$$4k = 14.00$$

$$k = 3.50$$

Each hour past the first 3 costs \$3.50, so the rule is $C(t) = 8.00 + 3.50(t - 3)$ for $t > 3$.

$$k = 3.50$$

6. Phone plan: \$30.00 base with 5 gigabytes included, \$12.00 per gigabyte above 5, bill capped at \$90.00. Find $C(9)$.

Three regions, so three branches — the cap is a branch of its own:

$$C(g) = \begin{cases} 30.00, & 0 \leq g \leq 5 \\ 30.00 + 12.00(g - 5), & 5 < g \leq 10 \\ 90.00, & g > 10 \end{cases}$$

At $g = 9$ the middle branch applies, and $9 - 5 = 4$ gigabytes are charged: $C(9) = 30.00 + 12.00 \cdot 4 = 30.00 + 48.00$ dollars, which is under the \$90.00 cap, so the cap does not bind.

$$C(9) = 78.00$$

7. Same plan: find the usage at which the middle branch first reaches the \$90.00 cap.

The cap takes over exactly where the middle branch's value hits \$90.00:

$$\begin{aligned} 30.00 + 12.00(g - 5) &= 90.00 \\ 12.00(g - 5) &= 60.00 \\ g - 5 &= 5 \\ g &= 10 \end{aligned}$$

So the middle branch runs on $5 < g \leq 10$ and the flat \$90.00 branch takes over for $g > 10$.

$$g = 10 \text{ GB}$$

8. Explain why the middle branch and the capped branch must agree where they meet. (*Open response — flagged for manual review.*)

Model answer: At $g = 10$ the middle branch gives $30.00 + 12.00(10 - 5) = 90.00$ dollars and the capped branch gives \$90.00. They agree, so the graph of C is one connected path with no vertical break. If they disagreed — say the cap were \$85.00 instead — then a customer using exactly 10 gigabytes would pay \$90.00 while a customer using 10.1 gigabytes would pay \$85.00. Using *more* data would lower the bill, so a customer could save money by streaming a video they did not want, and two customers whose usage differs by a rounding error would see bills \$5.00 apart. A fee schedule is only fair if cost never jumps down as usage goes up, which is exactly the requirement that the branches match at the breakpoint.

Grading note: full credit for any answer that (a) evaluates both branches at the breakpoint and observes they agree, and (b) describes the jump a customer would experience otherwise. Naming it “continuity” is welcome but not required at this level.

9. Garage A (\$4.00 up to 2 hours, then \$3.00 per hour) versus Garage B (\$2.50 per hour). Find the stay longer than 2 hours on which they cost the same.

For $t > 2$, Garage A costs $4.00 + 3.00(t - 2)$ and Garage B costs $2.50t$. Let y be the shared cost in dollars:

$$\begin{cases} y = 4.00 + 3.00(t - 2) \\ y = 2.50t \end{cases}$$

Substituting the second equation into the first:

$$2.50t = 4.00 + 3.00(t - 2)$$

$$2.50t = 3.00t - 2.00$$

$$2.00 = 0.50t$$

$$t = 4$$

Then $y = 2.50 \cdot 4 = 10.00$ dollars, and the check on Garage A gives $4.00 + 3.00(4 - 2) = 10.00$ dollars as well. The answer $t = 4$ lies in the $t > 2$ region the system was written for, so it is a legitimate solution.

$t = 4, y = 10.00$

10. Print shop: \$25.00 setup plus \$0.40 per page for the first 200 pages, then r dollars per page beyond 200; a 500-page job costs \$153.00. Find r .

$$C(p) = \begin{cases} 25.00 + 0.40p, & 0 \leq p \leq 200 \\ 25.00 + 0.40(200) + r(p - 200), & p > 200 \end{cases}$$

The first 200 pages always cost $25.00 + 0.40 \times 200 = 105.00$ dollars, and a 500-page job has $500 - 200 = 300$ pages at the reduced rate:

$$105.00 + r(500 - 200) = 153.00$$

$$300r = 48.00$$

$$r = 0.16$$

So the reduced rate is \$0.16 per page and the rule is $C(p) = 105.00 + 0.16(p - 200)$ for $p > 200$.

$r = 0.16$
